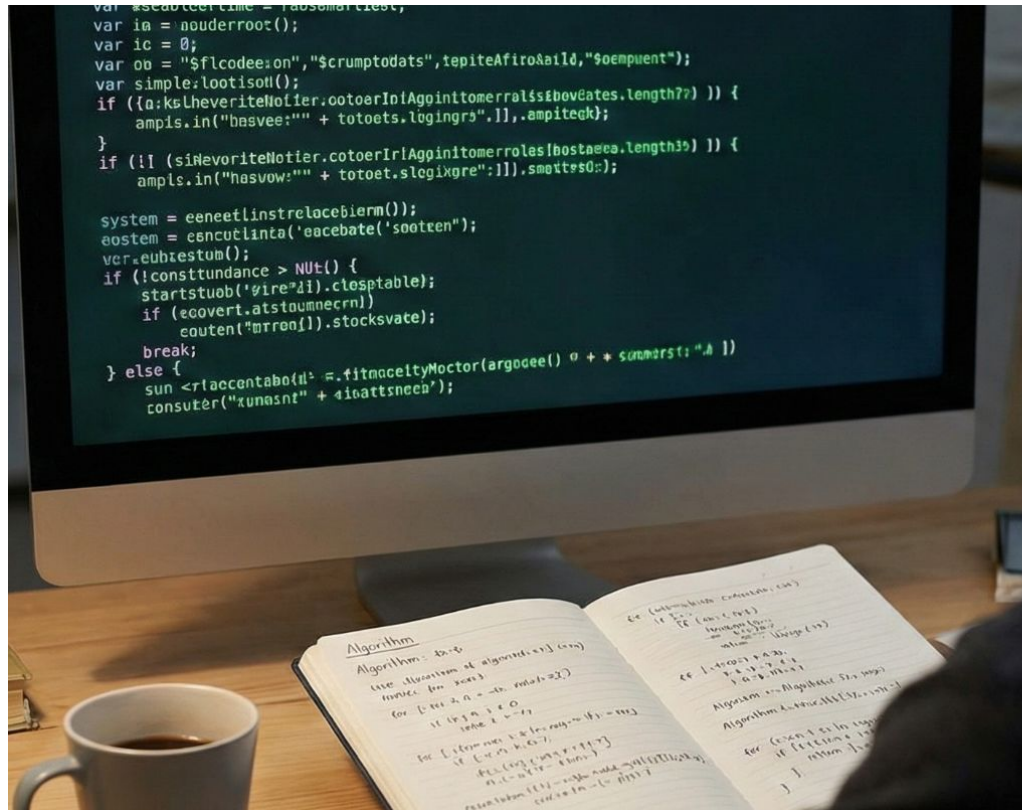


CSE470: Software Engineering

Introduction

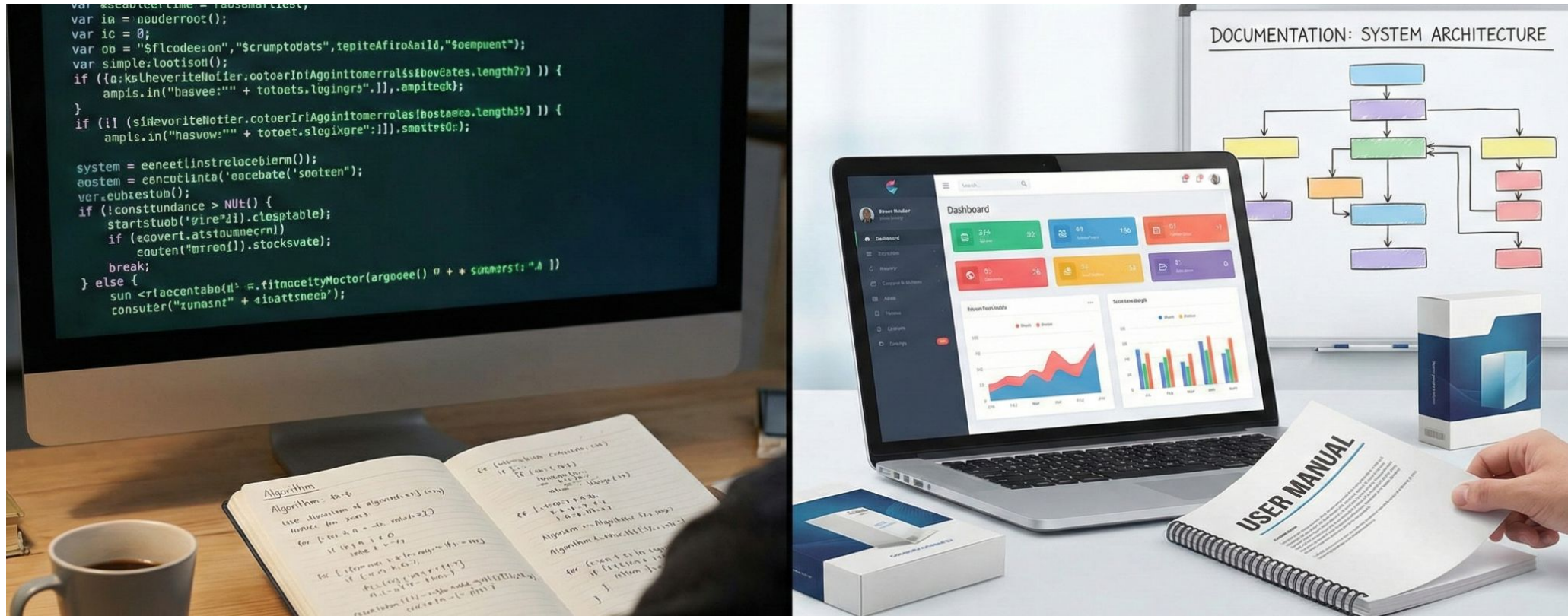


Program vs Software



Program vs Software

Software = Program + Documentation + User Guide



Software Crisis

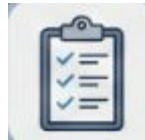
1. Scalability
2. Load Management
3. Maintain quality

Can be solved by Engineering

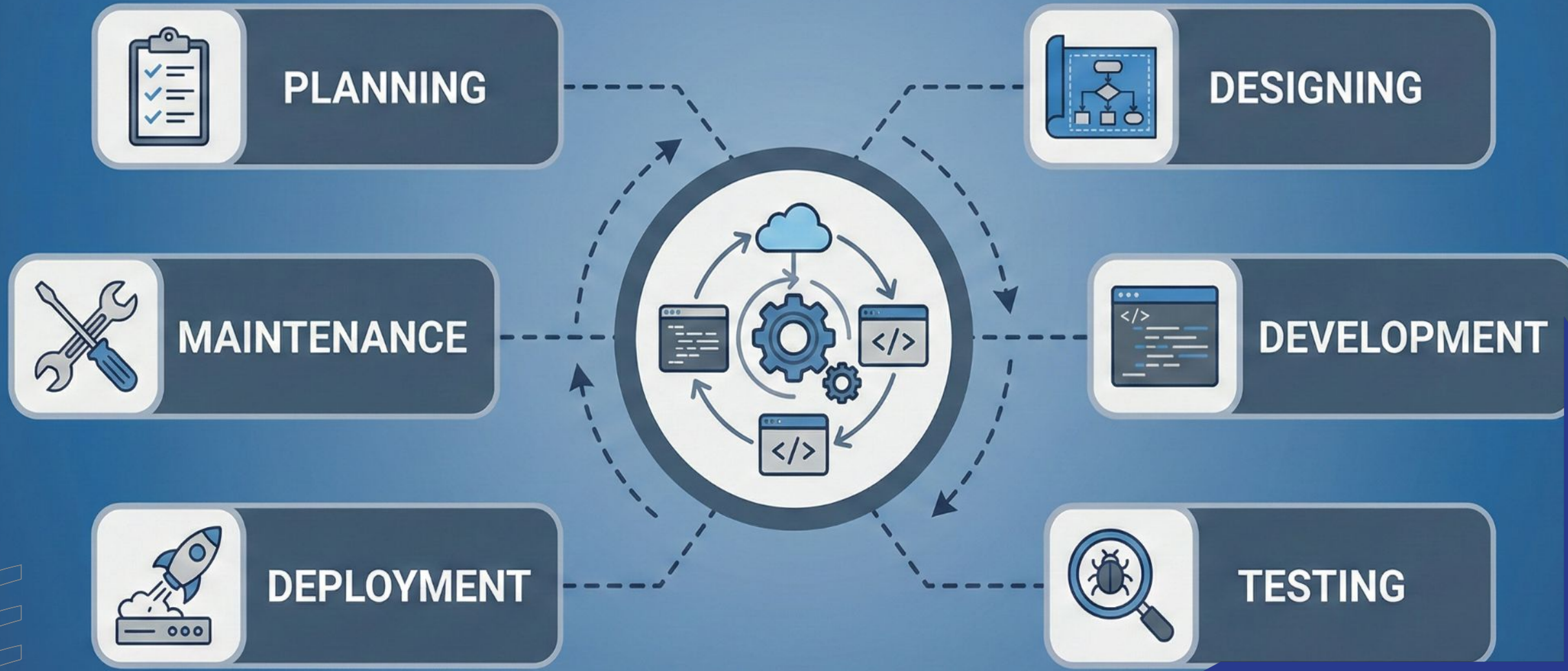


Software Engineering

What are the steps you need to build a software ?



Software Engineering



Requirement Analysis

In this phase, all the requirements are gathered from the client. Basically collecting information about what the client wants their software to do. Then our job is to:

1. Address the problem
2. Identify feasible and infeasible requirements
3. Identify how the software helps our client attain their requirements



Design

In this step, the requirements gathered in the previous step is converted into different format (**Diagrams**) that can be coded in a programming language.

- ERD, Schema Diagram, Class Diagram, Data Flow Diagram etc.

It usually includes a high level and detailed design and architecture of the proposed software solution.



Development/Coding

After the design is fixed:

1. The design is converted into blocks
2. Those blocks are converted into code
3. Step by step converting the design to a workable software



Testing

Testing phase starts after the development phase is completed.

- The finished product is tested based on the user requirements
- In case of any problem, it is fixed in code (Unit Testing)
- It is usually done in several phases



Deployment

- The software is deployed in the production environment
 - Host in the cloud server
 - Installation on client's server
 - Initial Setup
 - User manual / Training
- Test to check if the system is performing as expected



Maintenance

- If anything goes wrong in the deployment process, it is fixed here
- Keeping the software working over time



Model for Project Management

Waterfall Model



What is a Waterfall Model?

A **sequential** methodology for software project management.

There are times when the **requirements for a problem are well understood**—when work flows from communication through deployment in a reasonably **linear** fashion.

It is also called the **Classic Life Cycle**



When does it occur?

This situation is sometimes encountered when well-defined adaptations or enhancements to an existing system must be made

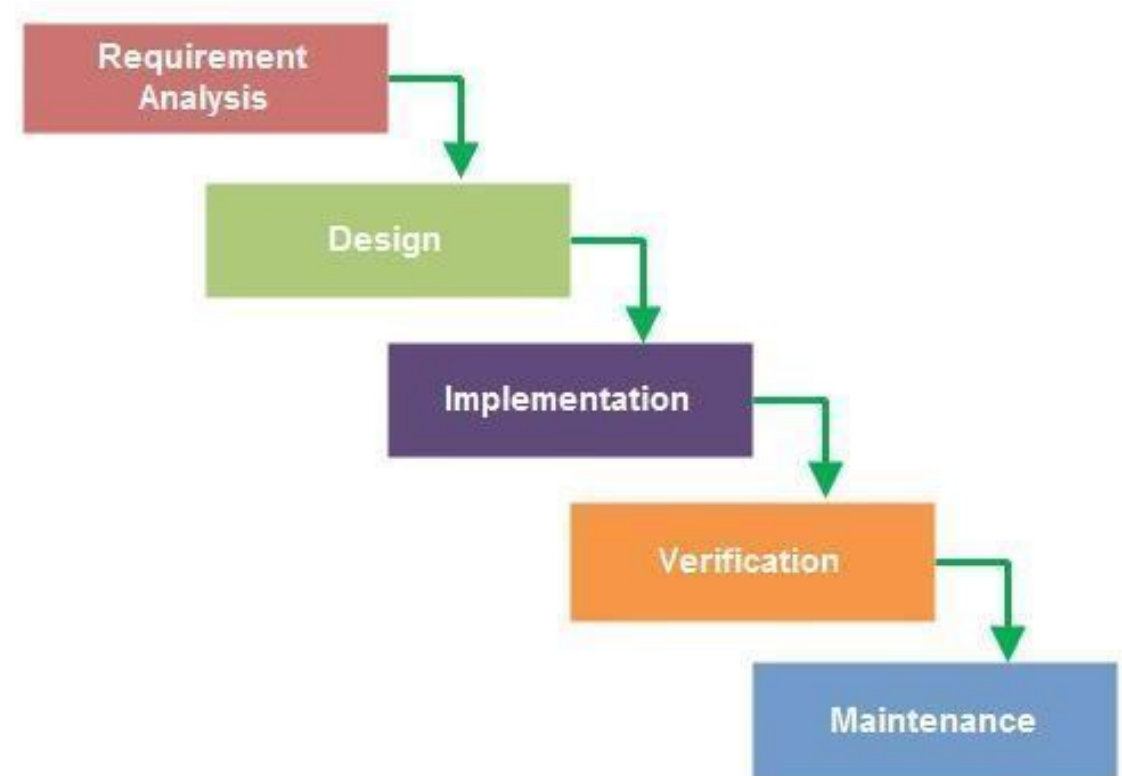
It may also occur in a limited number of new development efforts, but only when **requirements are well defined and reasonably stable**.

Examples:

- Creating a payroll system for a company where the rules for calculations are pre-defined and stable.
- Developing software for a medical device, like a pacemaker, where requirements are fixed by safety standards.
- Developing software for EVM, where strict legal protocols, security standards, and user interface requirements must be fully finalized and certified before deployment.

Steps of the Waterfall Model

1. Requirement Collection & Analysis
2. Design
3. Coding/Implementation
4. Testing/Verification
5. Deployment
6. Maintenance



Advantages of Waterfall Model

- Easy to Understand and Implement
- Each Step is properly defined
- Steps are performed one at a time
- Clear goals of each step



Disadvantages of Waterfall Model

- No feedback path
- Difficult to incorporate requirement changes
- No step overlapping
- Not flexible
- Limited Stakeholder Involvement
- Late error detection
- Working product after the last step of the model



When to choose Waterfall Model

- When Requirements are well known
- Little to no changes expected
- Small to Medium Projects
- Minimum Client Involvement needed
- Resources are available and trained

The Waterfall approach involves less user interaction in the product development process. The product can only be shown to end user when it is ready.



Case Study

Case 1: One of your uncle requested you to develop an accounting calculator for his local shop.

Case 2: Your start-up company wants to develop an accounting calculator for super shops.

Question: Determine if you can implement both cases in Waterfall Model



Case 1

One of your uncle requested you to develop an accounting calculator for his local shop.

Key Points to note:

- Local shop (Small scale)
- Already existing business (Fixed Requirement)

Verdict: Waterfall can be Implemented



Case 2

Your company wants to develop an accounting calculator for start-up super shops.

Key Points to note:

- Super shop (Big scale)
- Start up (Requirements are not fixed)

Verdict: Cannot implement Waterfall Model



References

1. Roger S Pressman, Roger Pressman, "Software Engineering: A Practitioner's Approach", McGraw-Hill, 7th edition, 2010.
2. <https://www.geeksforgeeks.org/waterfall-model/>